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Crack-Tip, Stress-Intensity Factors for Plane Extension and Plate Bending Problems

A complex variable method for evaluating the strength of stress singularities at crack tips in plane problems and plate bending problems is derived. The results of these evaluations give Irwin's stress-intensity factors for plane problems and analogous quantities for bending problems, a form familiar to the practitioner of "fracture mechanics." The methods derived are integrated with the complex variable approach of Muskhelishvili to obtain the stress-intensity factors for various basic examples applicable to the extension and bending of plates with through-the-thickness cracks. The results suggest the possibility of extension of the Griffith-Irwin fracture theory to arbitrary plane extensional and/or bending problems in plates.

THE concept of crack-tip, stress-intensity factors (which are in fact the strength of stress singularities at crack tips) applied to predictions of the static strength of cracked bodies has been developed by Irwin [1, 2, 3]² for plane extension, symmetric with respect to the crack. Stress-intensity factors have also been shown [4] to control the rate of crack propagation under cyclic loading in such situations. However, extension of these applications to other than symmetric plane problems and/or bending problems is yet to be accomplished.

Both Irwin [1, 2, 3] and Williams [5] have obtained the form of the elastic stress distribution in the vicinity of a crack tip in extensional problems. Williams [6], in a recent work, extends his analysis to thin plates subjected to bending out of the plane. In each case it is shown that the significant stresses in the vicinity

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² Numbers in brackets designate References at end of paper.

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of the crack tip are those associated with the singularity of stress of the order $(r)^{-1/2}$, where r is radial distance from the crack tip.

Moreover, the distribution of stress in each, extension or bending, is unique; i.e., its functional form in terms of co-ordinates measured from the crack tip is always the same. The intensity of the local field of stress may be represented in terms of two parameters. Irwin's stress-intensity factors are themselves a convenient form of these parameters which are now in use [4] for extensional problems. Hence, for bending, Williams' results will be modified to define stress-intensity factors in bending in a manner consistent with Irwin's definitions.

Then it will be shown that the complex variable technique of Muskhelishvili [7] and its extension to plate bending problems [8] can be conveniently incorporated into a new method for computing the stress-intensity factors for various configurations.

Stress Field Near Crack-Tip and Stress-Intensity Factors

In plane extension, the stress distribution in the vicinity of the end of a crack has been shown by Irwin to always take the form:

$$\sigma_x = \frac{k_1}{(2r)^{1/2}} \cos \frac{\theta}{2} \left[1 + \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right] + \frac{k_2}{(2r)^{1/2}} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \frac{3\theta}{2} \quad (\text{Cont.})$$

Nomenclature

- a = half crack length of a straight crack
 b = dimension locating a concentrated force on a crack surface
 b_m ($m = 1, 2$) = constants determined by loading and configuration
 D = flexural rigidity of a plate
 E = Young's modulus of elasticity
 F = concentrated extensional force on a crack surface (per unit thickness)
 G = shear modulus of elasticity
 h = thickness of a plate
 i = $(-1)^{1/2}$
 k_1, k_2 = crack tip stress-intensity factors for symmetrical and skew-symmetrical stress distribu-

tions, respectively, encountered in extensional problems

$$k = k_1 - ik_2$$

K_1, K_2 = crack tip stress-intensity factors for symmetrical and skew-symmetrical stress distributions, respectively, encountered in transverse bending problems

$$K = K_1 - iK_2$$

$M_r, M_\theta, M_{r\theta}$ = bending moments and twisting moment, respectively, all per unit length with reference to (r, θ) -directions

M_0, Q_0 = bending and shearing force per unit length, respectively, applied at a plate boundary.

P, Q = x and y -components of a force, F

(Continued on next page)

$$\sigma_x = \frac{k_1}{(2r)^{1/2}} \cos \frac{\theta}{2} \left[1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right] - \frac{k_2}{(2r)^{1/2}} \sin \frac{\theta}{2} \left[2 + \cos \frac{\theta}{2} \cos \frac{3\theta}{2} \right] \quad (1)$$

$$\tau_{xy} = \frac{k_1}{(2r)^{1/2}} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \frac{3\theta}{2} + \frac{k_2}{(2r)^{1/2}} \cos \frac{\theta}{2} \left[1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right]$$

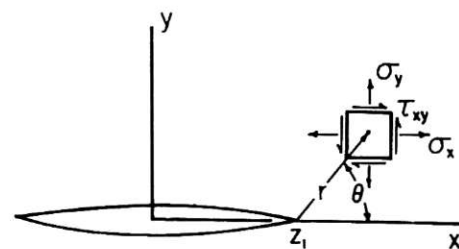


Fig. 1 Notation for rectangular stress components near crack tip

where higher-order terms in r have been neglected. Fig. 1 gives the notation used in this representation. The stress-intensity factors, k_1 and k_2 , are the parameters (dependent on the configuration and loading) which control the magnitude of the symmetric and skew-symmetric components of this stress distribution.

In the case of plate bending, Williams' results for the stresses are similar, i.e.,

$$\sigma_r = \frac{12M_r\delta}{h^3} = \frac{1}{r^{1/2}} \left\{ \left[\cos \frac{3\theta}{2} - \frac{3+5\nu}{7+\nu} \cos \frac{\theta}{2} \right] \frac{3b_1G\delta}{2} + \left[-\sin \frac{3\theta}{2} + \frac{3+5\nu}{5+3\nu} \sin \frac{\theta}{2} \right] \frac{3b_2G\delta}{2} \right\}$$

$$\sigma_\theta = \frac{12M_\theta\delta}{h^3} = \frac{1}{r^{1/2}} \left\{ \left[-\cos \frac{3\theta}{2} - \frac{5+3\nu}{7+\nu} \cos \frac{\theta}{2} \right] \frac{3b_1G\delta}{2} + \left[\sin \frac{3\theta}{2} + \sin \frac{\theta}{2} \right] \frac{3b_2G\delta}{2} \right\} \quad (2)$$

$$\tau_{r\theta} = \frac{12M_{r\theta}\delta}{h^3} = \frac{2}{r^{1/2}} \left\{ \left[-\sin \frac{3\theta}{2} + \frac{1-\nu}{7+\nu} \sin \frac{\theta}{2} \right] \frac{3b_1G\delta}{2} + \left[-\cos \frac{3\theta}{2} + \frac{1-\nu}{5+3\nu} \cos \frac{\theta}{2} \right] \frac{3b_2G\delta}{2} \right\}$$

where δ is a co-ordinate perpendicular to the middle plane of the plate, and again higher order terms in r have been neglected, Fig. 2.

For convenience, as previously mentioned, the constants, b_1 and b_2 , in Williams' expression, equations (2), are redefined by

$$K_1 = -\frac{3\sqrt{2}(3+\nu)Ghb_1}{7+\nu} \quad (3)$$

and

$$K_2 = -\frac{3\sqrt{2}(3+\nu)Ghb_2}{5+3\nu}$$

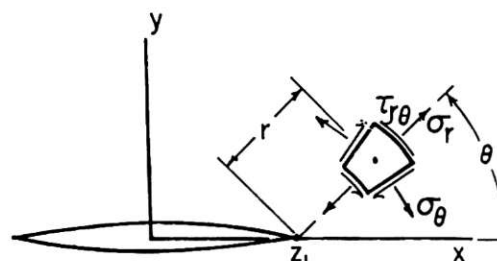


Fig. 2 Notation for polar stress components near crack tip

where K_1 and K_2 will henceforth be called the bending stress-intensity factors. The desirable character of this choice will be obviated as the analysis proceeds.

Now, for the purpose of subsequently developing methods for determining the stress-intensity factors for individual problems, the sum of the normal stresses (an invariant) will be useful. Forming these from equations (1), (2), and (3), they are, for extension:

$$\sigma_x + \sigma_y = k_1 \left[\frac{2}{r} \right]^{1/2} \cos \frac{\theta}{2} - k_2 \left[\frac{2}{r} \right]^{1/2} \sin \frac{\theta}{2} \quad (4)$$

and for bending:

$$\sigma_r + \sigma_\theta = \frac{12(M_r + M_\theta)\delta}{h^3} = \frac{2\sqrt{2}(1+\nu)\delta}{(3+\nu)h} \left[\frac{K_1}{\sqrt{r}} \cos \frac{\theta}{2} - \frac{K_2}{\sqrt{r}} \sin \frac{\theta}{2} \right]$$

In both cases the stress-intensity factors may be represented as the real and negative imaginary parts of a complex constant in accordance with

$$k = k_1 - ik_2 \quad (5)$$

$$K = K_1 - iK_2$$

Nomenclature

r, θ = polar co-ordinates measured from a crack tip and from line of extension of crack, respectively
 w = transverse deflection of a plate in bending, or Airy stress function for extension
 x, y = rectangular co-ordinates in middle plane of a plate
 z = complex variable ($x + iy$)
 α = half-angle subtended by a curved crack
 β = angle between direction of an applied shear force, Q_0 , or angle between a field of uniform extensional stress and crack plane
 δ = co-ordinate perpendicular to middle plane of a plate
 ζ = complex variable in a mapped plane

κ = elastic constant which takes values $\kappa = 3 - 4\nu$ for plane strain and $\kappa = (3 - \nu)/(1 + \nu)$ for plane stress
 ν = Poisson's ratio
 σ = uniform extensional stress at infinity
 $\sigma_r, \sigma_\theta, \tau_{r\theta}$ = bending-stress components with reference to (r, θ) -directions
 $\sigma_x, \sigma_y, \tau_{xy}$ = components of stress
 $\varphi(z), \psi(z), \chi(z), \varphi(\zeta)$ = Muskhelishvili's analytic stress functions of complex variables z and ζ , respectively
 φ_e, φ_b = Muskhelishvili's functions for extension and bending, respectively
 $\omega(\zeta)$ = a mapping function
 $\text{Re}\{ \}$ = real part of
 $'$, prime = derivative with respect to argument

Also introducing the complex variable

$$z = x + iy = z_1 + re^{i\theta}$$

where z_1 is the location of the crack tip, equations (4) may be written, for extension:

$$\sigma_x + \sigma_y = \operatorname{Re} \left\{ k \left[\frac{2}{z - z_1} \right]^{1/2} \right\}$$

and for bending,

$$M_r + M_\theta = \frac{(1 + \nu)h^2}{3\sqrt{2}(3 + \nu)} \operatorname{Re} \left\{ K \left[\frac{1}{z - z_1} \right]^{1/2} \right\} \quad (6)$$

The similarity of these results is evident; however, two points are to be kept in mind: (a) The stress fields from which they were derived are actually quite different (compare [5] and [6]), and (b), they are valid only in the immediate vicinity of the crack tip, as z approaches z_1 .

Application of Complex Variable Stress-Function Technique

A complex variable method of solving boundary-value problems governed by the biharmonic equation,

$$\nabla^4 w = 0 \quad (7)$$

has been developed principally by Muskhelishvili [7] (a résumé is presented in Sokolnikoff's work [9]). The application to many examples of extensional problems is treated in [7] and bending problems are given attention by Savin [8]. The biharmonic equation, equation (7), governs both extensional problems and bending problems (where no transverse loads are present). w is the Airy stress function for extension and the transverse deflection for bending.

In either case the general solution to equation (7) may be written

$$w = \operatorname{Re} \{ \bar{z}\varphi(z) + \chi(z) \} \quad (8)$$

where

$$\chi(z) = \int \psi(z) dz$$

Further, $\varphi(z)$ and $\psi(z)$ are analytic functions of the complex variable z . The solution to an individual extension or bending problem is reduced to finding the functions, $\varphi(z)$ and $\psi(z)$, which satisfy the boundary conditions of that problem.

Without going into the details of determining $\varphi(z)$ and $\psi(z)$, a method of determining stress-intensity factors is to be developed. Hence it is expedient to only state the well-known [7, 8] relationships between the stress functions and the quantities in the left-hand members of equations (6). They are, for extension,

$$\sigma_x + \sigma_y = 4 \operatorname{Re} \{ \varphi_e'(z) \} \quad (9)$$

and for bending,

$$M_r + M_\theta = -4D(1 + \nu) \operatorname{Re} \{ \varphi_b'(z) \}$$

where $D = Eh^3/[12(1 - \nu^2)]$ is the flexural rigidity of the plate.

In the vicinity of a crack tip $\varphi'(z)$ takes the form of a complex constant divided by $(z - z_1)^{1/2}$, reference [7] (pp. 496-498). Therefore, equating the results of equations (6) and (9) and recalling that equations (6) are valid only as z approaches z_1 , the stress-intensity factors may be evaluated from, for extension,

$$k = 2\sqrt{2} \lim_{z \rightarrow z_1} (z - z_1)^{1/2} \varphi_e'(z) \quad (10)$$

and for bending,

$$K = -\frac{12\sqrt{2} D(3 + \nu)}{h^2} \lim_{z \rightarrow z_1} (z - z_1)^{1/2} \varphi_b'(z)$$

Now, equations (10) illustrate the fact that the stress-intensity factors may be found simply from a knowledge of $\varphi'(z)$ in the vicinity of the crack tip. Other methods can be derived for like determinations; for example, expansions for the stress or moments in the vicinity of the crack tip in a particular problem yield the stress-intensity factors directly. However, it is the authors' opinion that it is usually much easier to perform the indicated operation of equations (10) than to compute and expand the stress for the multitude of problems where the functions $\varphi(z)$ and $\psi(z)$ are already known [7-13]. Moreover, in solving new problems it is somewhat easier to simply find $\varphi(z)$ and ignore $\psi(z)$ if stress-intensity factors are the only desired result. The power of the technique of equations (10) is best illustrated by the simplicity of providing examples for stress-intensity factors heretofore not available in the literature.

Curved Crack in Sheet Under Biaxial Tension

Fig. 3 shows the problem of an infinite sheet under uniform biaxial tension with a circular-sector crack of unit radius about the origin. For this problem Muskhelishvili [7] gives the derivative of the stress function,

$$\varphi_e'(z) = \frac{\sigma}{2 \left(1 + \sin^2 \frac{\alpha}{2} \right)} \left[\frac{z - \cos \alpha}{(1 - 2z \cos \alpha + z^2)^{1/2}} + \sin^2 \frac{\alpha}{2} \right] \quad (11)$$

In order to treat the problem by making use of equations (10), the co-ordinates must first be rotated such that the crack tip of interest is parallel to the x -axis. A convenient transformation for this purpose (which also relocates the crack tip on the x -axis) is

$$z = ie^{i\alpha}(\bar{z} - i - \sin \alpha \cos \alpha) \quad (12)$$

Substituting equation (12) into (11) and the result subsequently into the first of equations (10) gives

$$k_1 = \frac{\sigma}{1 + \sin^2 \frac{\alpha}{2}} \left[\frac{\sin \alpha (1 + \cos \alpha)}{2} \right]^{1/2} \quad (13)$$

and

$$k_2 = \frac{\sigma}{1 + \sin^2 \frac{\alpha}{2}} \left[\frac{\sin \alpha (1 - \cos \alpha)}{2} \right]^{1/2}$$

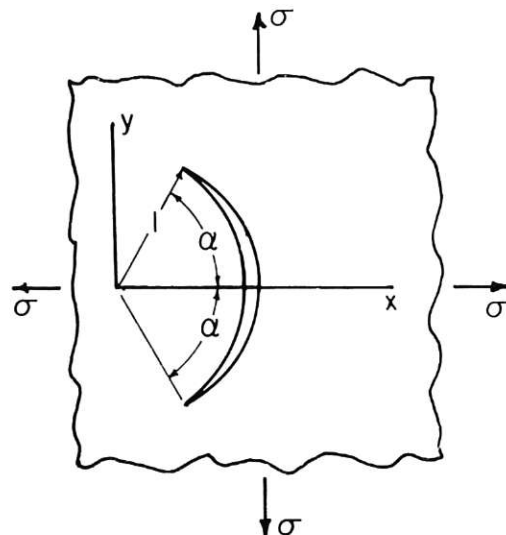


Fig. 3 Infinite sheet containing curved crack of unit radius

Conformal Mapping

In many extensional and bending problems the use of conformal mapping is an expedient. Let the mapping function be defined (with the usual restrictions as to analyticity and single-valuedness) by

$$z = \omega(\zeta) \quad (14)$$

Then

$$\varphi'(z) = \frac{d\varphi}{dz} \cdot \frac{d\zeta}{dz} = \frac{\varphi'(\zeta)}{\omega'(\zeta)} \quad (15)$$

Now, corresponding to the crack tip z_1 in the z -plane there will be a point ζ_1 in the ζ -plane. Thus equations (10) may be written, for extension:

$$k = 2\sqrt{2} \lim_{\zeta \rightarrow \zeta_1} [\omega(\zeta) - \omega(\zeta_1)]^{1/2} \frac{\varphi'_e(\zeta)}{\omega'(\zeta)} \quad (16)$$

and for bending:

$$K = -\frac{12\sqrt{2} D(3+\nu)}{h^2} \lim_{\zeta \rightarrow \zeta_1} [\omega(\zeta) - \omega(\zeta_1)]^{1/2} \frac{\varphi'_b(\zeta)}{\omega'(\zeta)}$$

Furthermore, for mappings which transform the "sharp" crack tip into a smooth curve an additional simplification results. For example, the mapping,

$$z = \omega(\zeta) = \frac{a}{2} \left(\zeta + \frac{1}{\zeta} \right) \quad (17)$$

transforms a straight crack of length, $2a$, in the x -direction, located centrally at the origin in the z -plane, into a circular hole of unit radius in the ζ -plane. Upon noting that the point corresponding to the crack tip, $z = a$, is at $\zeta = 1$ in the ζ -plane, equations (16) and (17) may be combined to give, for extension,

$$k = \frac{2}{a^{1/2}} \varphi'_e(\zeta)|_{\zeta=1} \quad (18)$$

and for bending:

$$K = -\frac{12D(3+\nu)}{h^2 a^{1/2}} \varphi'_b(\zeta)|_{\zeta=1}$$

since with a circular boundary present $\varphi'(\zeta)$ has no singularity at $\zeta = 1$. Equations (18) will be used to find the stress-intensity factors in the examples to follow.

Concentrated Extensional Force on Crack Surface

The extensional problem of an infinite sheet with a crack along the x -axis of length, $2a$, centered at the origin and having an arbitrary extensional force (per unit sheet thickness), F , applied to its upper surface at $x = b$ is of special interest, since it may be used as a Green's function to form the solution to other problems. For this problem, Fig. 4, the stress function is

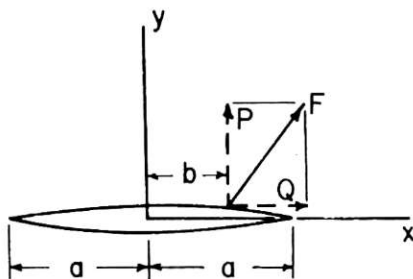


Fig. 4 Crack whose surface is subjected to concentrated force

$$\varphi_e(\zeta) = \frac{Fa}{2\pi(a^2 - b^2)^{1/2}} \left\{ -\frac{1}{\zeta} + \left(\frac{\zeta_0}{\zeta_0 - \zeta} \right) \left[\left(\zeta + \frac{1}{\zeta} \right) - \left(\zeta_0 + \frac{1}{\zeta_0} \right) \right] + \left(\zeta_0 + \frac{1}{\zeta_0} \right) \left[\frac{\kappa}{1+\kappa} \log \zeta - \log(\zeta_0 - \zeta) \right] \right\} \quad (19)$$

where the crack has been mapped into a unit circle, by equation (17), on which $\zeta = \zeta_0$ corresponds to $x = b$ on the upper surface of the crack. The concentrated force may be expressed in terms of its y and x -components, P and Q , respectively, whereupon

$$F = P - iQ \quad (20)$$

Using the first of equations (18) to evaluate the stress-intensity factors (at the right end of the crack):

$$k = \frac{P - iQ}{2\pi a^{1/2}} \left[\left(\frac{a+b}{a-b} \right)^{1/2} + i \left(\frac{\kappa-1}{\kappa+1} \right) \right]$$

$$\text{or} \quad k_1 = \frac{P}{2\pi a^{1/2}} \left(\frac{a+b}{a-b} \right)^{1/2} + \frac{Q}{2\pi a^{1/2}} \left(\frac{\kappa-1}{\kappa+1} \right) \quad (21)$$

$$\text{and} \quad k_2 = -\frac{P}{2\pi a^{1/2}} \left(\frac{\kappa-1}{\kappa+1} \right) + \frac{Q}{2\pi a^{1/2}} \left(\frac{a+b}{a-b} \right)^{1/2}$$

The use of equations (21) is self-evident in handling problems of an infinite sheet with a crack on whose surface tractions are applied.

In addition, problems of an infinite sheet with a crack whose surface is free from tractions may be attacked by superposition and resolved to the solution of the problem of a crack with surface tractions. This is accomplished by solving the original problem for the stresses at the crack site, $\sigma_y(x, 0)$ and $\tau_{xy}(x, 0)$, with the crack absent and then superimposing tractions equal and opposite to the crack-site stresses. The result is

$$k_1 = \frac{1}{\pi a^{1/2}} \int_{-a}^a \sigma_y(x, 0) \left(\frac{a+x}{a-x} \right)^{1/2} dx$$

$$k_2 = \frac{1}{\pi a^{1/2}} \int_{-a}^a \tau_{xy}(x, 0) \left(\frac{a+x}{a-x} \right)^{1/2} dx \quad (22)$$

Equations (22) represent a simple direct method of determining the stress-intensity factors for a crack in an infinite plate with arbitrary extensional loading.

Example of Bending of Infinite Plate Containing Crack

Consider the problem of an infinite plate containing a crack of length $2a$, where a bending moment, M_0 , is applied all around the boundary of the plate at infinity, Fig. 5. Savin [8] gives the Muskhelishvili function for this problem using equation (17):

$$\varphi_b(\zeta) = -\frac{M_0 a}{4D(1+\nu)} \left[\zeta + \frac{1-\nu}{3+\nu} \frac{1}{\zeta} \right] \quad (23)$$

Introducing equation (23) into the second of equations (18) results in

$$K_1 = \frac{6M_0}{h^2} a^{1/2} \quad (24)$$

$$K_2 = 0$$

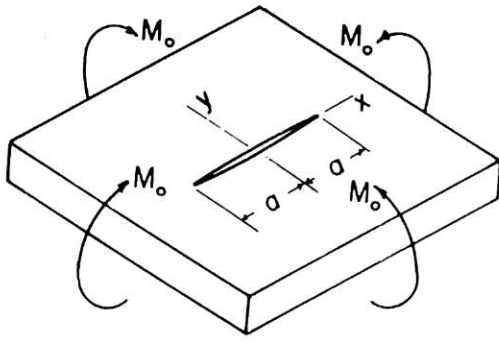


Fig. 5 Infinite plate containing crack and subjected to uniform bending

Upon observing that $6M_0/h^2$ is the stress in the surface layer of the plate at infinity, this result is similar to the "corresponding" extensional problem; i.e., uniform extensional stress, σ , at infinity which gives [1] $k_1 = \sigma a^{1/2}$. The convenient simplicity and familiarity of this result are in part the justification for defining the bending stress intensity factors as in equations (3). Similar simplicity and analogies with extension problems occur in other bending problems, as will be seen in the discussions to follow.

The three preceding examples of determination of stress-intensity factors, equations (13), (21), and (24), are sufficient to illustrate the techniques developed here, equations (10) and (18). It is easy to add more examples; therefore, for the use of the engineer an Appendix is attached to this work giving some additional results in compact form. It is now more pertinent to discuss the significance of these new results with regard to their possible applications.

Extension of Griffith-Irwin Theory for Thin Sheet Structures

Thin plates (and shells with radii of curvature which are very large compared to the crack length) which contain a crack are often the subject of inquiries on their remaining strength. The aircraft industry is one example of interest in such problems. To date, the Griffith-Irwin theory has enjoyed a unique amount of success in treating many of these problems. Yet, so far it has been almost entirely restricted in application to plane extensional problems, symmetric with respect to the crack.

An example of a more general problem is shown in Fig. 6. This element could be a portion of a sheet in a structure, where the introduction of the crack will have little influence on the loading forces if the element is large in planar dimensions compared to the crack length. Presuming the loading is known from a structural analysis, the stress-intensity factors may be found by superposition of those for simpler individual loadings. The stress-intensity factors for each of the individual loadings is given in the Appendix.

Hence, for this example, Fig. 6,

$$\begin{aligned} k_1 &= \frac{N_y}{h} a^{1/2} \\ k_2 &= \frac{N_{xy}}{h} a^{1/2} \\ K_1 &= \frac{6M_y}{h^2} a^{1/2} + \frac{8Q_x a^{1/2}}{h^2} \\ K_2 &= \frac{6M_{xy}}{h^2} a^{1/2} + \frac{8Q_y a^{1/2}}{h^2} \end{aligned} \quad (25)$$

The simple extension of the Griffith-Irwin theory to such a problem is:

The combination of stress-intensity factors present will cause

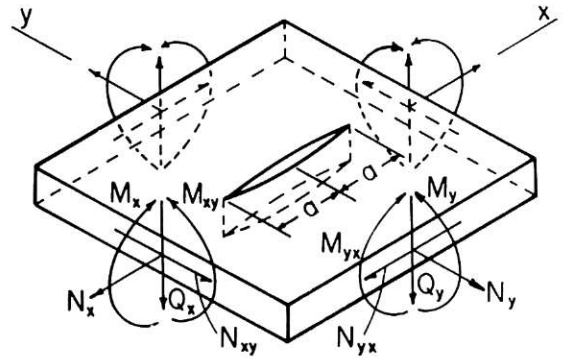


Fig. 6 Element containing crack removed from sheet structure

unstable crack extension upon reaching some critical value; i.e., the criterion takes the form

$$f(k_1, k_2, K_1, K_2) = f_{cr} \quad (26)$$

However, some doubtful assumptions remain. For example, interference between crack surfaces has been ignored. Moreover, with bending present, crack extension is likely to favor one side of the sheet more than the other.

Therefore this work is offered as a step toward more general application of Griffith-Irwin concepts, but it should be recognized that practical application must be accompanied by validation or modification of some of the assumptions present.

In addition, it is possible that such a generalization of the Griffith-Irwin concepts may carry over into the field of crack extension under cyclic loading [4]. Current studies are underway in an attempt to apply these ideas.

Conclusions

The complex variable approach presented gives a convenient way to determine crack-tip, stress-intensity factors for plane extension and plate bending problems, provided the Muskhelishvili stress function, $\phi(z)$, is known in either the actual or a mapped plane. Since Muskhelishvili's methods are well known for their power and direct approach for multiply connected regions, their incorporation is a natural one in analyzing cracks.

Examples have been given of the computation of stress-intensity factors for a few particular problems, which illustrate the power of the technique presented. These examples and other results are tabulated in the Appendix for the use of the engineer in fracture analyses. Thus the most important result may not necessarily be the technique of determining stress-intensity factors, but the primary contribution here is perhaps that the way is cleared for more general application of the Griffith-Irwin fracture concepts as indicated.

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References

- 1 G. R. Irwin, "Analysis of Stresses and Strains Near the End of a Crack Traversing a Plate," *JOURNAL OF APPLIED MECHANICS*, vol. 24, TRANS. ASME, vol. 79, 1957, pp. 361-364.
- 2 G. R. Irwin, "Fracture," *Handbuch der Physik*, vol. 6, Springer, 1958, pp. 551-590.
- 3 G. R. Irwin, "Fracture Mechanics," *Structural Mechanics*, Pergamon Press, New York, N. Y., 1960, pp. 557-592.

4 W. E. Anderson and P. C. Paris, "Fracture Mechanics Applied to the Evaluation of Aircraft Material Performance," *Metals Engineering Quarterly*, vol. 1, 1961, pp. 33-44.

5 M. L. Williams, "On the Stress Distribution at the Base of a Stationary Crack," *JOURNAL OF APPLIED MECHANICS*, vol. 24, TRANS. ASME, vol. 79, 1957, pp. 109-114.

6 M. L. Williams, "The Bending Stress Distribution at the Base of a Stationary Crack," *JOURNAL OF APPLIED MECHANICS*, vol. 28, TRANS. ASME, Series E, vol. 82, March, 1961, pp. 78-82.

7 N. I. Muskhelishvili, "Some Basic Problems of Mathematical Theory of Elasticity," 1933, English translation, P. Noordhoff and Company, New York, N. Y., 1953.

8 G. N. Savin, "Spannungserhöhung am Rande von Löchern," VEB Verlag Technik, Berlin, Germany, 1956, German translation.

9 I. S. Sokolnikoff, "Mathematical Theory of Elasticity," McGraw-Hill Book Company, Inc., New York, N. Y., 1956, pp. 288-289.

10 J. L. Sanders, Jr., "On the Griffith-Irwin Fracture Theory," *JOURNAL OF APPLIED MECHANICS*, vol. 27, TRANS. ASME, Series E, vol. 82, 1960, pp. 352.

11 Yi-Yuan Yu, "The Influence of a Small Hole or Rigid Inclusion on the Transverse Flexure of Thin Plates," *Proceedings of Second U. S. National Congress of Applied Mechanics*, 1954, pp. 381-387.

12 M. Goland, "The Influence of the Shape and Rigidity of an Elastic Inclusion on the Transverse Flexure of Thin Plates," *JOURNAL OF APPLIED MECHANICS*, vol. 10, TRANS. ASME, vol. 65, 1943, p. A-69.

13 J. N. Goodier, "The Influence of Circular and Elliptical Holes on the Transverse Flexure of Elastic Plates," *Philosophical Magazine*, series 7, vol. 22, 1936, pp. 68-80.

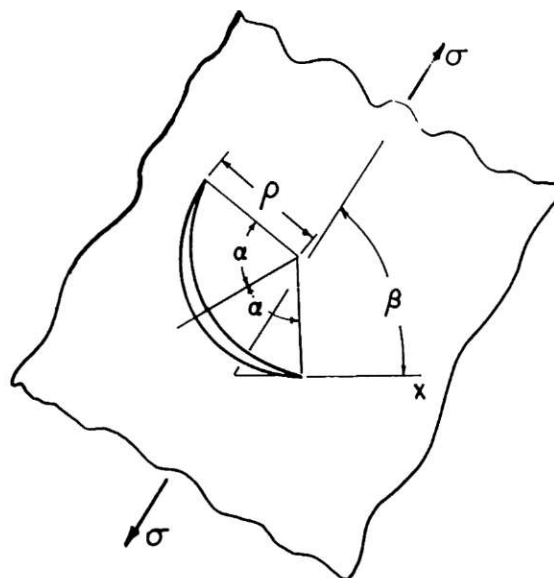


Fig. 8

Reference [7]

$$k_1 = \frac{\sigma(\rho \sin \alpha)^{1/2}}{2 \left(1 + \sin^2 \frac{\alpha}{2} \right)} \left\{ \cos \frac{\alpha}{2} + \cos \left(2\beta + \frac{5}{2} \alpha \right) \left[\sin^2 \frac{\alpha}{2} \right] \right. \\ \left. - \cos \left(2\beta + \frac{3}{2} \alpha \right) \left[\cos^2 \frac{\alpha}{2} - \sin^4 \frac{\alpha}{2} \right] \right. \\ \left. - \sin \left(2\beta + \frac{3}{2} \alpha \right) \left[\sin \alpha \sin^2 \frac{\alpha}{2} \right] \right\} \\ k_2 = \frac{\sigma(\rho \sin \alpha)^{1/2}}{2 \left(1 + \sin^2 \frac{\alpha}{2} \right)} \left\{ \sin \frac{\alpha}{2} + \sin \left(2\beta + \frac{5}{2} \alpha \right) \left[\sin^2 \frac{\alpha}{2} \right] \right. \\ \left. + \sin \left(2\beta + \frac{3}{2} \alpha \right) \left[\cos^2 \frac{\alpha}{2} - \sin^4 \frac{\alpha}{2} \right] \right. \\ \left. - \cos \left(2\beta + \frac{3}{2} \alpha \right) \left[\sin \alpha \sin^2 \frac{\alpha}{2} \right] \right\}$$

Infinite Plates Subjected to Bending (and Shearing)

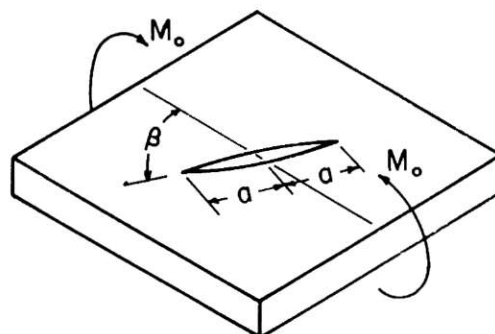


Fig. 9

Reference [8]

$$\varphi_b = -\frac{M_o(\sin^2 \beta)a}{4D(1 + \nu)} \left[\zeta + \frac{1 - \nu}{3 + \nu} \frac{1}{\zeta} \right] - \frac{iM_o(\sin 2\beta)a}{4D(3 + \nu)} \left[\frac{1}{\zeta} \right]$$

APPENDIX

The following tabulation of results is presented to supply the reader with the stress-intensity factors for various basic configurations and loadings not available in previous literature. For the sake of brevity, the few examples chosen will be those which can be easily modified and/or superimposed to give solutions to a variety of special case problems which may be of engineering significance. With each result a single reference will be listed where more information on that example may be obtained.

Infinite Plates Subjected to Extensional Loading

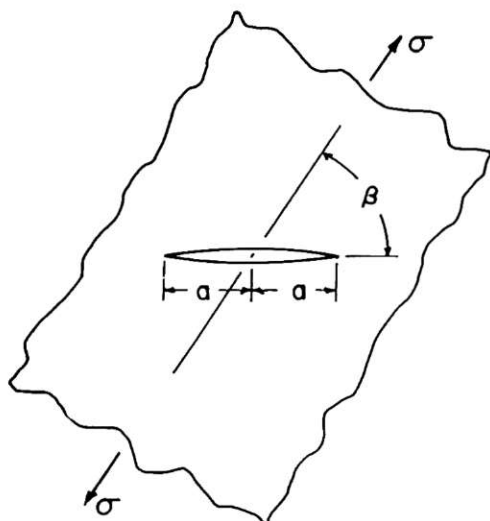


Fig. 7

Reference [7]

$$\varphi_e = -\frac{\sigma(1 - e^{2i\beta})a}{4\zeta} \\ k_1 = \sigma(\sin^2 \beta)a^{1/2} \\ k_2 = \sigma(\sin \beta \cos \beta)a^{1/2}$$

$$K_1 = \frac{6M_0}{h^2} (\sin^2 \beta) a^{1/2}$$

$$K_2 = \frac{6M_0}{h^2} (\sin \beta \cos \beta) a^{1/2}$$

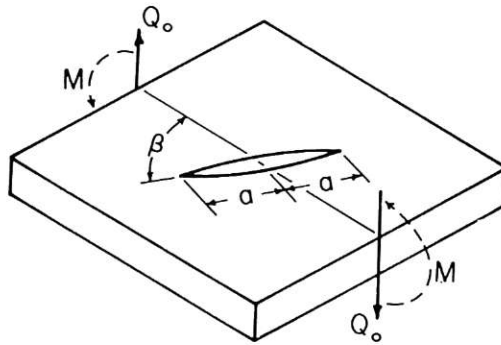


Fig. 10

Reference [11]

$$\varphi_b = -\frac{Q_0 a^2}{12D} \left[\zeta^2 e^{-i\beta} + \left(\frac{1-\nu}{3+\nu} \right) \frac{e^{-i\beta} + 2e^{i\beta}}{\zeta^2} \right]$$

$$K_1 = \frac{8\nu Q_0 a^{3/2}}{h^2} \cos \beta$$

$$K_2 = \frac{8Q_0 a^{3/2}}{h^2} \sin \beta$$